

Surendra Kumar Chandrasekaran

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EDUCATION

University of Maryland – College Park

M.S. in Applied Machine Learning

Relevant Coursework: Principles of Machine Learning, Data Science, Probability and Statistics

Sep 2025 – May 2027

GPA: 4.0/4.0

College of Engineering, Guindy (Anna University)

B.E. in Computer Science and Engineering

Relevant Coursework: Data Structures and Algorithms, Computer Architecture, Operating Systems, DBMS

Nov 2021 – May 2025

GPA: 3.7/4.0

SKILLS

Languages: Python, C/C++, Java, SQL, JAX

ML/AI: PyTorch, HuggingFace, scikit-learn, Transformers, NLP, Computer Vision, RAG, Model Optimization

LLM/Agents: LangGraph, LangChain, LlamaIndex, MCP, Agentic Workflows, Tool Calling, Prompt Engineering

Engineering: FastAPI, Docker, AWS, GCP, Linux, Git, Redis

Data: PostgreSQL, MySQL, MongoDB, FAISS, PySpark, NumPy, Pandas

EXPERIENCE

Software Developer Intern – fusionSpan LLC

Jun 2026 – Aug 2026

- Engineered an autonomous LLM-based debugging agent that ingests Jenkins CI logs, identifies failing tests, applies targeted fixes, and uses bounded retry loops to reduce manual debugging time for recurring failures by 50%.
- Automated generation of monthly site reports to be sent to clients using the WordPress MCP and Google Analytics 4 Data API and integrated both with a custom HTML template to generate each report that removed the overhead of manual intervention.

Graduate Research Assistant – University of Maryland

Sep 2025 – Present

- Designed a RAG pipeline for political discourse analysis from a variety of podcast transcripts using FAISS and compared multiple LLM inference evals to identify the best suitable one for the task.
- Developed scalable multimodal evaluation pipelines combining text and computer vision models to analyze coded language patterns and detect cross-modal signals.

Machine Learning Intern – Kaar Technologies

May 2024 – Feb 2025

- Designed a multi-document QA RAG pipelines using LangChain, FAISS, and TinyLLaMA-1.1B, reducing query latency by 35% through retrieval and inference optimization.
- Built a lightweight KPI retrieval system using Word2Vec and spaCy for semantic extraction and classification of performance indicators, reducing processing time by 40%.

PROJECTS

ALTA: Adaptive Logistics and Tracking Agent

Apr 2026 – May 2026

- Orchestrated a real-time agentic logistics system using LangGraph, FastAPI, React, and LLM-based risk scoring, with multi-agent orchestration and conditional DAG workflows for shipment monitoring that decreased latency by ~50%
- Deployed the application with SQLite persistence and human-in-the-loop workflows, enabling automated risk assessment and intervention across concurrent cold-chain shipments.

CaseSage: Analysis of Indian Legal Documents

Nov 2024 – May 2025

- Devised a Legal NLP system for Indian legal judgments using Longformer-based summarization and NER to extract and organize information from lengthy court documents.
- Implemented knowledge-graph-based precedent retrieval and RAG-based QA, improving legal information retrieval and reducing lookup time for case analysis.

AgentLens: AI Agent Evaluation & Observability Platform

Jan 2026 – Apr 2026

- Architected an open-source AI agent evaluation platform using LangGraph, FastAPI, PostgreSQL, Redis, and OpenTelemetry to trace LLM calls, tool usage, retrieval, and multi-step agent workflows.
- Established automated benchmarks for task success, tool-selection accuracy, retrieval quality, hallucination, latency, and token efficiency, with an MCP-based server for trace inspection and agent failure analysis.

AWARDS

Gemini Hack Night

Nov 2025

- Secured 1st place among 150+ participants by building an AR-based CPR assistance web application in 5 hours using Google MediaPipe, Gemini Pro, and ReactJS, achieving 92% measurement accuracy.

CERTIFICATIONS

Claude Certified Architect – Foundations

July 2026

- Validated expertise in Agentic Architecture, MCP Tool Design, Context Engineering, Structured-Output Generation and Claude Code configuration for Production-Grade Claude Deployments.